

HangerBody: a Haptic Device Using Haptic Illusion for Multiple Parts of Body

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Figure 1: (a)Hanger Reflex, (b)HangerBody device, which presents force sensation to head, elbow, wrist, and knee, (c) Force feedback in VR fighting game, (d)Force feedback in VR swing training, (e) Assistance with force sensation in the reaching task rehabilitation, (f)Haptic feedback in remote assistance

ABSTRACT

The hanger reflex is a phenomenon in which a strong sense of rotational force is perceived when one wears a wire hanger on his/her head. This phenomenon is caused by lateral skin stretch generated by pressure from the wire hanger on the head, and similar phenomena have been observed in the wrist, ankle, knee and elbow. However, the hanger reflex has been applied to only one part of the body, but not to multiple parts of the body simultaneously. In this study, we propose the HangerBody system, which applies the hanger reflex to multiple parts of body simultaneously in order to expand the range of applications. In this paper, we introduce the system configuration and some examples of applying the system to VR games, sports training, rehabilitation, and remote assistance.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

Force Feedback, Whole Body Haptics, Hanger Reflex

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1 INTRODUCTION

The hanger reflex is a phenomenon in which a strong sense of rotational force is perceived when one wears a wire hanger on his/her head[Sato et al. 2009](Figure 1(a)). This phenomenon is caused by lateral skin stretch generated by pressure from the wire hanger on the head[Miyakami et al. 2018], and similar phenomena have been observed in the wrist, ankle, and knee[Kon et al. 2016][Sanada et al. 2020]. The phenomenon is controlled by reproducing the pressure pattern of the hanger reflex with actuators[Sato et al. 2009]. Compared with conventional haptic devices that present physical force, this phenomenon can present force sensation with a simple and compact device, because this phenomenon presents perceptual force, not physical force. Particularly, the hanger reflex is more powerful than conventional methods using perception or illusion such as asymmetric acceleration vibration[Rekimoto 2014] or skin deformation[Yem et al. 2015]. This phenomenon has already been applied to force feedback in walking navigation and VR experiences[Kon et al. 2017]. However, it has been applied only to one part of the body, but not to the multiple parts of the body simultaneously.

In this study, we propose the HangerBody system, which can apply the hanger reflex to the multiple parts of body simultaneously(Figure 1(b)) to expand the range of applications. In this paper, we introduce the system configuration and some examples of applying the system to VR games, sports training, rehabilitation, and remote assistance.

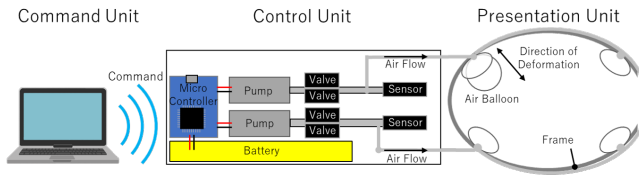


Figure 2: System configuration of HangerBody

2 SYSTEM OF HANGERBODY

Following the previous study[Kon et al. 2017], the HangerBody system consists of a command unit, a control unit, and a presentation unit that are attached to each body part(Figure 2).

The command unit consists of a motion capture system for measuring body movements, measurement markers, and a PC for computation. The user wears a measurement marker, and the motion capture system measures the user's movements. The measured information is sent to the PC for computation, and the simulator in the PC judges the interaction. When a collision or a command from the controller is detected, a drive command is sent to the control unit.

The control unit controls the amount of air supplied to the presentation unit and the pressure pattern based on the command from the command unit. The control unit consists of a small air pump, a solenoid valve, an air pressure sensor, a microcomputer with a wireless communication module, and a battery. The microcomputer receives commands from the control unit and supplies air to the balloons in the presentation unit through air pumps and valves based on the commands. The amount of air supplied is determined based on the value of the air pressure sensor connected to each balloon, and the appropriate air pressure is maintained. The maximum number of connected air balloons is 28 (4 air balloons/1 presentation unit x 7 parts), and this unit controls the amount of air supplied to all air balloons.

The presentation unit consists of a frame corresponding to the mounting part of body and four air balloons, and reproduces the compression during the hanger reflex by adjusting the inflation of the balloons. By changing the pressure pattern, the direction of the force sensation presented is changed. The frame corresponding to the wearing area is made of 3D-printed ABS, and the multiple sizes of frames will be prepared to fit most of the users. This unit can be worn at a maximum of seven parts: right and left elbows, right and left wrists, right and left knees, and head.

This section describes the driving method of the presentation unit. Previous studies have shown that the hanger reflex occurs at the head, wrist, and knee. In this study, we newly applied the hanger reflex to the elbow, and confirmed that a similar phenomenon occurs. The direction of the force sensation to be presented is determined by the direction of the lateral skin stretch generated by the compression of the air balloon[Miyakami et al. 2018]. In this paper, four air balloons are installed in the presentation unit, so that force perception can be presented in one axis of rotation and two axes of translation (Figure 3).

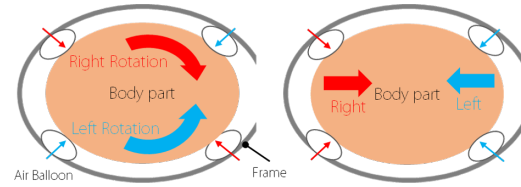


Figure 3: (Left)Pressure patterns for rotational force, (Right)Pressure patterns for translational force

3 DEMO EXPERIENCE

In order to demonstrate the potential of this system, we will use the device on seven parts of the body and provide force feedback to VR game(Figure 1(c)). In this demo, we use the device to feedback the sense of force when attacking or being attacked when playing a VR fighting game. It enables force feedback not only to the head but also to the knees and elbows to enrich the game experience. In VR sports training, force feedback is presented to the head, elbows, wrists, and knees in conjunction with swinging movements to support the acquisition of appropriate swinging motion(Figure 1(d)). This system can also be applied to rehabilitation support and remote assistance(Figure 1(e)(f)). In the rehabilitation application, the system supports the posture and movement of the arm by presenting force feedback to the elbow and wrist during reaching. For remote assistance, a remote helper can guide the posture of the operator to realize smooth teleoperation. Conversely, the system can also be used to provide force feedback to the helper on collisions and resistance to manipulation during task.

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REFERENCES

- Yuki Kon, Takuto Nakamura, and Hiroyuki Kajimoto. 2017. HangerON: A Belt-Type Human Walking Controller Using the Hanger Reflex Haptic Illusion. In *ACM SIGGRAPH 2017 Emerging Technologies* (Los Angeles, California) (SIGGRAPH '17). Association for Computing Machinery, New York, NY, USA, Article 10, 2 pages. <https://doi.org/10.1145/3084822.3084833>
- Yuki Kon, Takuto Nakamura, Michi Sato, and Hiroyuki Kajimoto. 2016. Effect of Hanger Reflex on walking. In *2016 IEEE Haptics Symposium (HAPTICS)*. 313–318. <https://doi.org/10.1109/HAPTICS.2016.7463195>
- Masahiro Miyakami, Yuki Kon, Takuto Nakamura, and Hiroyuki Kajimoto. 2018. Optimization of the hanger reflex (I): examining the correlation between skin deformation and illusion intensity. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*. Springer, 36–48.
- Jun Rekimoto. 2014. Traxion: A Tactile Interaction Device with Virtual Force Sensation. In *ACM SIGGRAPH 2014 Emerging Technologies* (Vancouver, Canada) (SIGGRAPH '14). Association for Computing Machinery, New York, NY, USA, Article 25, 1 pages. <https://doi.org/10.1145/2614066.2614079>
- Hanamichi Kanada, Masato Kobayashi, Yuki Kon, and Hiroyuki Kajimoto. 2020. Investigation of Effective Parts for Rotation and Translation of the Legs Using Hanger Reflex. In *Proceedings of the Augmented Humans International Conference* (Kaiserslautern, Germany) (AHs '20). Association for Computing Machinery, New York, NY, USA, Article 14, 6 pages. <https://doi.org/10.1145/3384657.3384786>
- Michi Sato, Rika Matsue, Yuki Hashimoto, and Hiroyuki Kajimoto. 2009. Development of a head rotation interface by using Hanger Reflex. In *RO-MAN 2009 - The 18th IEEE International Symposium on Robot and Human Interactive Communication*. 534–538. <https://doi.org/10.1109/ROMAN.2009.5326327>
- Vibol Yem, Mai Otsuki, and Hideaki Kuzuoka. 2015. Development of wearable outer-covering haptic display using ball effector for hand motion guidance. In *Haptic Interaction*. Springer, 85–89.